

Satellite to Ground FSO Communication Links enhanced by Mode Diversity Reception and Adaptive Optics

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Abstract We experimentally demonstrate a joint adaptive optics and mode diversity reception (AO+MDR) scheme for GEO satellite-to-ground laser communication. A 1 Gbit/s BPSK link achieves 4.0 dB power gain and improves the probability of BER $< 1 \times 10^{-3}$ from 54.2% to 88.8%, confirming synergistic compensation. © 2026 The Author(s)

Introduction

Satellite laser communication offers high data rates, large bandwidth, strong confidentiality, and strong resistance to electromagnetic interference, making it attractive for global connectivity, efficient information transmission, and secure data delivery. In particular, satellite-to-ground downlinks are essential for massive data return and space science exploration [1]. However, the performance of satellite-to-ground optical links is severely limited by atmospheric effects. Atmospheric turbulence induces random refractive-index fluctuations, resulting in wavefront distortion and intensity scintillation, which cause severe received-power fluctuations and degrade communication reliability [2]. Therefore, mitigating atmospheric effects is essential for realizing high-capacity satellite-to-ground laser communication with low interruption probability. Existing turbulence mitigation approaches mainly include aperture diversity, multiple-input multiple-output techniques, adaptive optics (AO), and mode diversity reception (MDR).

Among existing turbulence mitigation approaches, AO is a mainstream solution because it actively corrects wavefront distortion to improve beam quality and fiber coupling performance. Its effectiveness has been demonstrated in several satellite-to-ground laser communication experiments. The Laser Communications Relay Demonstration program deployed an AO system at a California ground station and achieved a Strehl ratio of 0.7 and a coupling efficiency of 55% under turbulence with $r_0 = 5$ cm at $\lambda = 550$ nm [3]. The European Space Agency reported effective coupling enhancement for GEO and LEO coherent optical links using AO under strong turbulence with $r_0 = 5$ cm at $\lambda = 1064$ nm [4]. In addition, a 1.76 m telescope equipped with a high-precision AO system enabled 10 Gbit/s satellite-to-ground laser communication with the *Shijian-20* satellite,

improving the single-mode fiber coupling power from about -70 dBm to about -45 dBm under favorable turbulence conditions [5]. Another AO-assisted satellite-to-ground experiment with the *Jilin-1* MF02A04 satellite also achieved 10 Gbit/s transmission [6]. Despite these advances, practical AO deployment in GEO downlinks remains challenging. High coupling efficiency requires high-order correction, high-speed wavefront sensing, and high-bandwidth real-time control, resulting in substantial system complexity and cost [7]. In addition, under low-elevation and strong-scintillation conditions, conventional AO performance can be degraded by system delay, measurement loss, and rapidly varying turbulence [8].

MDR is another effective approach for turbulence mitigation. By exploiting multiple orthogonal modes supported by few-mode or multimode fibers as parallel channels, MDR enables more reliable signal recovery from multiple received replicas. Its practical potential has also been demonstrated in a recent 100 Gbit/s LEO satellite-to-ground laser communication experiment. In parallel, extensive studies have investigated MDR for free-space optical links. Ref [9] established simulation models for LEO satellite-to-ground links with different mode-diversity schemes, showing that both few-mode and multimode photonic lanterns can provide clear power and signal-to-noise ratio gains over single-mode reception, with multimode photonic lanterns offering superior performance. Ref [10] proposed an MDR receiver based on a spatial demultiplexer and a silicon photonic coherent combiner, and numerically verified its collection efficiency for GEO satellite-to-ground links in a laboratory setting.

Although both AO and MDR have shown effective turbulence mitigation, existing GEO satellite-to-ground demonstrations still mainly rely on AO alone. Compared with LEO links, GEO downlinks are more challenging because of the much

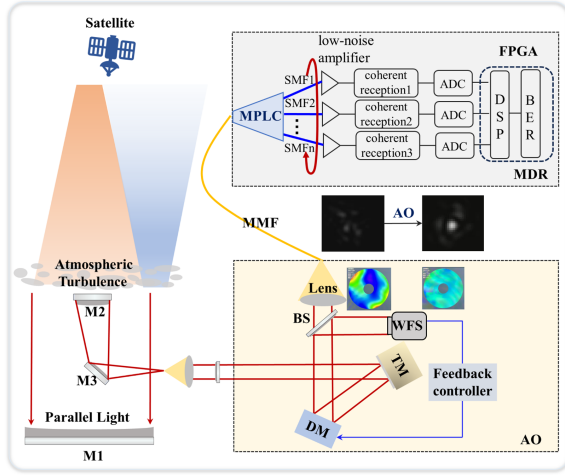


Fig. 1: Experimental setup of the GEO satellite-to-ground laser communication system.

longer propagation distance and higher reception loss. In this context, AO+MDR is a promising solution for high-capacity GEO satellite-to-ground laser communication with low interruption probability. Our previous 15.6 km horizontal free-space experiment demonstrated the significant compensation advantage of AO+MDR under strong turbulence [11]. This paper presents a GEO satellite-to-ground laser communication experiment employing AO+MDR compensation. Compared with AO-only compensation, the proposed scheme improves the received power by 4.0 dB at a complementary cumulative distribution function (CCDF) of 0.99, increases the zero-error-frame probability by 34.6%, and enables stable transmission of a 1 Gbit/s BPSK signal, confirming the feasibility and effectiveness of AO+MDR for GEO downlinks.

Experimental Setup

The GEO satellite-to-ground laser communication setup is shown in Fig. 1. The experiment was conducted with a 1.76 m telescope over a 36,705 km link, with an onboard transmit power of 2 W. The system consisted of the satellite terminal, the ground telescope, the Coudé AO system, and the communication receiver. The satellite terminal performed signal transmission, acquisition, and tracking, while the ground station handled downlink reception, acquisition, and tracking, as well as uplink off-axis transmission for uplink-downlink isolation. After link establishment, the received beam was guided into the Coudé optical path, which integrated tracking, AO correction, and fiber coupling. At the receiver, the AO+MDR module was employed for turbulence mitigation. The beam was first corrected by a 357-element AO system and then split by a 9:1 beamsplitter, with 10% sent to a Shack-Hartmann wavefront sensor and 90% coupled into the fiber. The coupled signal was subsequently mode-demultiplexed by a multi-plane light converter (MPLC) with an insertion loss of 1.2 dB

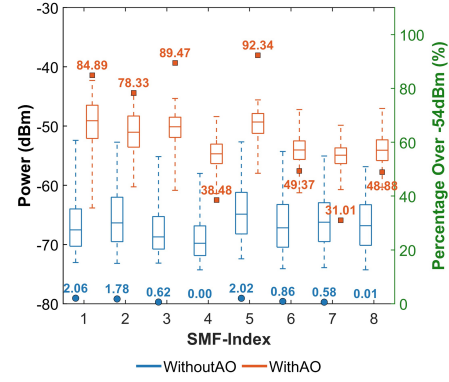


Fig. 2: Boxplot statistics of MPLC single-mode output powers and the probability of exceeding the sensitivity threshold.

to generate a single-mode array output, followed by coherent reception and signal combining.

Statistical Characterization of Power

During stable link operation, the atmospheric coherence length (r_0) referred to the telescope entrance pupil was estimated from the aperture wavefront slopes using the differential image motion method. The average r_0 was 7.43 cm, indicating a moderate-to-strong turbulence regime. Meanwhile, the output powers of the eight single-mode MPLC channels were recorded at 10 kHz for 10 s, the corresponding boxplot statistics and the probabilities of exceeding the sensitivity threshold of -54 dBm are shown on the left and right y-axes of Fig. 2, respectively. The upper and lower whiskers in the figure denote the power levels at CCDF = 0.01 and CCDF = 0.99, respectively. The experimental results show that the received powers of all single-mode outputs of the MPLC are significantly increased after AO correction, and the probabilities of exceeding the sensitivity threshold are also markedly improved. Among them, Channels 5, 3, and 1 exhibit the largest increases, reaching 92.3%, 89.5%, and 84.9%, respectively. This indicates that, under the current turbulence condition, the multimode-end modes corresponding to these channels account for relatively larger fractions of the distorted optical field. Using -54 dBm as the reference sensitivity, the best N channels ($N = 1, 2, \dots, 8$) were selected according to the number of samples exceeding this threshold for selection combining (SC) and maximum-ratio combining (MRC), as shown in Fig. 3. The corresponding boxplot statistics are shown in Fig. 3. Without AO compensation, the MPLC single-mode outputs remain very weak: even with 8-channel SC and MRC, the output powers at CCDF = 0.99 are still below -70 dBm and -64 dBm, respectively. For both SC and MRC, the combined output power increases with the number of modes, but the gain from SC is limited, whereas that from MRC is much more pronounced. When the number of combined modes reaches 3–5, the power gain tends to saturate. For example, as the num-

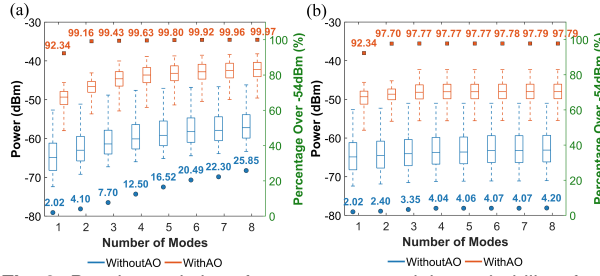


Fig. 3: Boxplot statistics of output power and the probability of exceeding the sensitivity threshold for (a) MRC and (b) SC.

ber of modes increases from 1 to 3, the SC and MRC gains are 0.7 dB and 1.3 dB, respectively, under AO open-loop operation, but increase to 4.0 dB and 6.9 dB under AO closed-loop operation. This clearly shows that AO enhances MDR performance. Conversely, MDR further enhances AO: under AO closed-loop operation, the power at CCDF = 0.99 increases from -74.36 dBm to -65.16 dBm. When the number of modes increases to 2, 3, and 4, the MRC output power improves from -65.16 dBm to -61.32 dBm, -58.23 dBm, and -57.24 dBm, corresponding to additional gains of 3.84, 6.93, and 7.92 dB beyond the 9.2 dB AO gain of the fundamental-mode channel. By contrast, the SC gain saturates at about 4.0 dB for three or more modes. These results verify that MDR can further strengthen AO. The right y-axis indicates the probability that the SC- and MRC-combined output powers of the best N channels exceed the sensitivity threshold of -54 dBm. When the number of modes increases from 1 to 2, AO correction improves this probability from 92.3% to 99.2%. Overall, the results demonstrate a mutually reinforcing effect between AO and MDR in the AO+MDR scheme. When AO correction is incomplete, residual higher-order modal energy can still be efficiently collected by MDR, improving the combined output power. Conversely, AO concentrates modal energy into lower-order modes, which further enhances MDR performance. This confirms the strong complementarity of AO and MDR for turbulence mitigation.

Statistical Characterization of BER

A 1 Gbit/s BPSK transmission experiment was carried out using a three-branch coherent receiver. During the experiment, the best three single-mode outputs of the MPLC were selected for coherent detection and analog-to-digital conversion. Under AO closed-loop operation, more than 1.22×10^5 frames of data per second were processed in real time on an FPGA, and the bit-error rate (BER) was acquired online through an Ethernet interface. The BER was determined by counting the number of erroneous bits in each frame containing 8192 bits.

Under AO closed-loop operation, the probability distributions of the pre-forward-error-correction

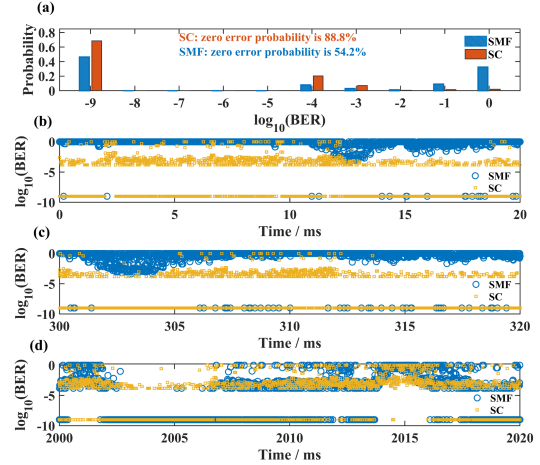


Fig. 4: (a) Probability distribution of BER and (b)–(d) time-varying BER curves over three representative intervals.

(pre-FEC) BER for the fundamental-mode signal and the signal after SC were compared over a 5 s interval during stable link operation, as shown in Fig. 4(a). Here, “SMF” and “SC” denote the pre-FEC BER of the fundamental-mode channel and the SC-combined signal, respectively. When the BER was below 1×10^{-3} , the frame was regarded as error-free, whereas a BER above 1×10^{-3} indicated that the frame could not be recovered using standard coding techniques. The results show that, under AO closed-loop operation, the zero-error-frame probabilities of the single-channel fundamental-mode signal and the SC signal were 54.2% and 88.8%, respectively. Therefore, SC increases the zero-error-frame probability by 34.6%. Figs. 4(b)–(d) further show the frame-by-frame BER evolution over three representative 20 ms intervals. It can be seen that the switching operation of the SC algorithm significantly reduces the BER and effectively avoids communication degradation caused by extremely low received power. By rapidly switching to the more stable mode, SC improves both the reliability and stability of data transmission, further confirming the effectiveness of SC-based MDR under complex propagation conditions. Therefore, compared with single-mode reception, MDR with SC provides better signal quality, and the AO+MDR scheme contributes to a more stable communication link.

Conclusions

This paper investigated an AO+MDR turbulence compensation scheme for GEO satellite-to-ground laser communication. An on-orbit experiment achieved stable transmission of a 1 Gbit/s BPSK signal. Compared with AO-only compensation, the proposed scheme improves the received power by 4.0 dB at CCDF = 0.99 and increases the probability of BER below 1×10^{-3} from 54.2% to 88.8%. These results confirm that SC-based MDR outperforms single-mode reception and that joint AO and MDR can effectively improve link stability.

Acknowledgements

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